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Thermal design and management towards high capacity CO_2 adsorption systems

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ABSTRACT

One of the most immediate technologies to mitigate the emission of carbon dioxide is carbon capture and storage. This study investigates different thermal design alternatives to increase the capacity of adsorption system given a desired cycle time. This is equivalent to investigating the reduction of the cooling and heating times during the adsorption and desorption processes, respectively. The thermal design investigation involves the variation of bed aspect ratio, the option of using water as a coolant, and augmenting the bed by fins with varying length and pitch. Mg-MOF-74 is used as adsorbent since it shows high CO_2 uptake at flue gas conditions. An experimentally validated CFD model is implemented by using user-defined-function (UDF) linked to the ANSYS Fluent program. Results show that rising the bed aspect ratio from 2.8 to 41.6 improves the CO_2 uptake by 13% and reduces the desorption period to the third. Additional improvement in the CO_2 uptake (10.9–13.6%) can be obtained by cooling and heating the bed by water as a function of the bed aspect ratio. The improvement jumps to 22.2% if fins are used to augment the heat transfer rates in the bed. The optimal CO_2 uptake enhancement (35.2%) is achieved by using finned annular-pipe (fin-pitch is 12.5 mm) as the adsorbent bed with water cooling. This design methodology can be adopted to other adsorption systems, including solar-driven adsorption chillers and solar-driven adsorption desalination systems.

1. Introduction

Nowadays, global warming is known to be one of the most critical environmental issues that threaten life on Earth. Greenhouse gases (GHG), including carbon dioxide, water vapor, nitrogen oxide, and methane, are the primary contributors to this issue. Among them, carbon dioxide is well documented to be the most gas of GHG affecting global warming. The excessive combustion of fossil material such as coal, natural gas, and oil is the source of carbon dioxide emission. However, fossil fuels are the dominant source of energy and will continue for the next few decades [1]. Accordingly, carbon capture sequestration (CCS) is obligatory for the continuity of using fossil fuels while reducing carbon dioxide emission, thereby mitigating global warming.

Many methods could be utilized for CCS, such as using porous material, membrane-based separation, or cryogenic. The utilization of porous materials has been receiving much attention because of its advantages of low cost and low regeneration energy requirements [2]. It can be used for different adsorbents, which could be cheap for the usage of large-scale post-combustion technologies [3]. The most common porous material used for the CO_2 adsorption process is activated carbon, silica gel, and zeolite, which have been studied widely from different viewpoints [4–16]. Lately, a new class of porous materials has been synthesized, the so-called metal-organic frameworks (MOFs) [17–24]. MOF-74 is widely being used since it has high adsorption capacity for carbon dioxide and excellent selectivity for CO_2 over N_2 [25,26]. Mg-MOF-74 is known to be one of the best adsorbents for carbon dioxide, where it can reach up to 350 mg g^{-1} at 298 K and 1 bar [27]. Yao et al. [28] showed that Mg-MOF-74 particles have high intrinsic selectivity for CO_2 adsorption over CH_4 and N_2. Oliveira et al. [29] performed density functional theory calculations for M-MOF-74 (M = Mg, Co, or Mn), where they indicated that this kind of calculations can assist the design of new optimized materials and even to recognize processes already described in the literature.

The heat transfer in adsorbent beds has attracted the researcher's attention due to its significant impact on gas uptakes and separation processes. Solmus et al. [30] investigated several design parameters that affect CO_2 uptake on silica gel adsorbent by using a one-

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Nomenclature

Abbreviations

CCS	Carbon capture and sequestration
CFD	Computational fluid dynamic
LDF	Linear driving force
UDF	User-defined function

Symbols

g	Gravity acceleration (m s^2)
h	Enthalpy (J/kg)
h_{ext}	Heat transfer coefficient of external environment (W m^{-2} K^{-1})
k	Thermal conductivity (W m^{-1} K^{-1})
k_L	Adsorption time constant (s^{-1})
p	Pressure (Pa)
v	Velocity (m s^{-1})
q	Amount of the gas adsorbed (mol of gas/kg of adsorbent)
t	Time (s)
y	Mass fraction of the gas species
C	Concentration (mol m^{-3})
C_0	Initial concentration (mol m^{-3})
C_2	Inertia resistance (m^{-1})
D	Diffusion coefficient (m^2 s^{-1})
D_o	External bed diameter *(m)*
E_g and E_s	Total gas energy and total adsorbent energy (J)
J	Diffusion flux (kg m^{-2} s^{-1})
M	Molecular weight (kg/kmol)
Nu	Nusselt number
Pr	Prandtl number
Ra	Rayleigh number
$S_{momentum}$	Source term in the momentum.
T	Temperature (K)
R	Universal gas constant (J mol^{-1} K^{-1})

Greek symbols

ρ	Density (kg m^{-3})
$\bar{\bar{\tau}}_{ij}$	Stress tensor, N m^{-2}
ε	Bed porosity
μ	Dynamic viscosity (Pa s)
$1/\kappa$	Viscous resistance of the porous media (m^{-2})

Subscript

b	Bed
g	Gas
i	Gas species (CO_2 or N_2)
m	Maximum
p	Particle
s	Solid adsorbent material

dimensional local thermal non-equilibrium model. They suggested reducing the interior mass transfer resistances rather than the exterior one. Further development of their model to be a two-dimensional model [31] which showed that less than 1 cm bed thickness is proper to increase heat transfer in the bed. Many literature studies focused on adsorbent properties that showed a low thermal conductivity, which varies between 0.044 and 0.188 W m^{-1} K^{-1} [32]. Therefore, it is required either to enhance the heat and mass transfer properties of adsorbents or to design novel adsorbent beds having better heat transfer dissipation. Demir et al. [33] improved the effective thermal conductivity and diffusivity of the silica gel by 157% and 242%, respectively, by adding 15 %wt aluminum to the adsorbent material. Also, they studied the effect of porosity on the heat and mass transfer in a granular adsorbent bed of silica gel. The increase in bed porosity leads to increasing the adsorption period. Caglar [34] studied the performance of a finned tube-type adsorbent bed filled by silica gel. The study found that doubling the fin thickness has lightly impact on the heat transfer rates while expanding the fin radius reduces the temperature by 10–17 K. Kowsari et al. [35] compared the performance of finned adsorption heat exchangers that have a rectangular and trapezoidal shape. They concluded that the rectangular bed is more appropriate since the specific cooling power is higher than in the case of the trapezoidal bed, especially at a shorter length of the bed. Zhao et al. [36] applied the concept of finned-tube in a solar adsorption refrigeration system using activated carbon/methanol as a working pair. The presence of fins resulted in more uniform temperature distribution through the activated carbon and hence reduced the possibility of methanol reabsorption and heat losses compared to those of the smooth tube. Furthermore, a cylindrical bed with annular fins is investigated by Niazmand and Dabzadeh [37] for adsorption chiller applications. It showed that applying the fins can reduce the bed size dramatically, but it also has a negative impact on the coefficient of performance (COP). Another study identified that the COP rises as the number of fins increases until a specific value, after which the COP remains approximately stable [38].

In the present work, we investigate the improvement in the adsorption capacity of adsorption systems by understanding the detailed dynamic behavior of the adsorption and desorption processes using a developed transient multi-dimensional mathematical model; and choosing different thermal design alternatives to reach our goal. The thermal design tools used include. This includes the use of fins with different length and pitch, inserting cooling water channels and varying the aspect ratio of the adsorption bed. Mg-MOF-74 is chosen as a CO_2 adsorbent, while the mathematical model is implemented using User-Defined-Functions (UDFs) in combination with ANSYS Fluent program for two-dimensional time-dependent modeling.

2. Adsorption/desorption system description

The adsorption/desorption cycle is described in Fig. 1. The gas mixture enters the adsorption bed and passes through the adsorbent material, which selectively adsorbs carbon dioxide over other gases until the bed gets saturated with CO_2. The flue gas is then redirected, and the bed is heated to a higher temperature to reject a particular amount of adsorbed CO_2. After that, the remaining CO_2 is purged out using N_2. The bed is then cooled and prepared for the next adsorption cycle [39].

Fig. 2a describes the adsorption bed of the base case. It consists of a circular tube made of stainless steel and filled with an adsorbent (e.g., Mg-MOF74). The feeding gas mixture (CO_2 and N_2) enters the adsorption bed from the left side; then, it moves through the adsorbent, which adsorbs CO_2 gas from the mixture while N_2 leaves the bed from the other side.

The adsorption process is known to be an exothermic reaction process; therefore, eight cooling configurations are proposed in order to reduce generated heat from the adsorption process. These configurations can be classified into two categories; the first one has a circular pipe passing through the center of the bed, as shown in Fig. 2c. The pipe carries water for bed cooling. In the second category, an annular pipe is used to cool the adsorbent bed. It is located in such a way that it divides the bed volume equally, as shown in Fig. 2b. For the two methods of cooling (i.e., circular pipe and annular pipe), fins are attached to the

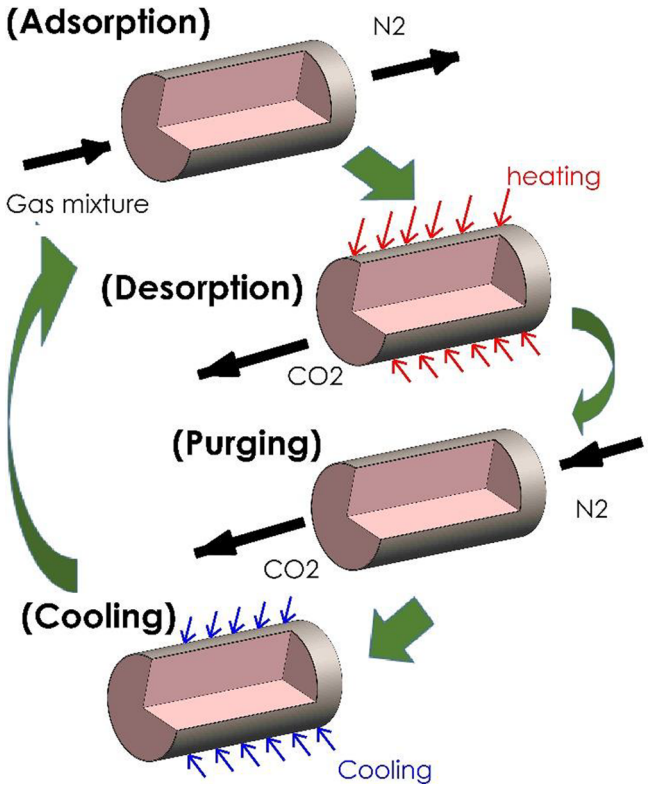


Fig. 1. Schematic of the adsorption/desorption cycle.

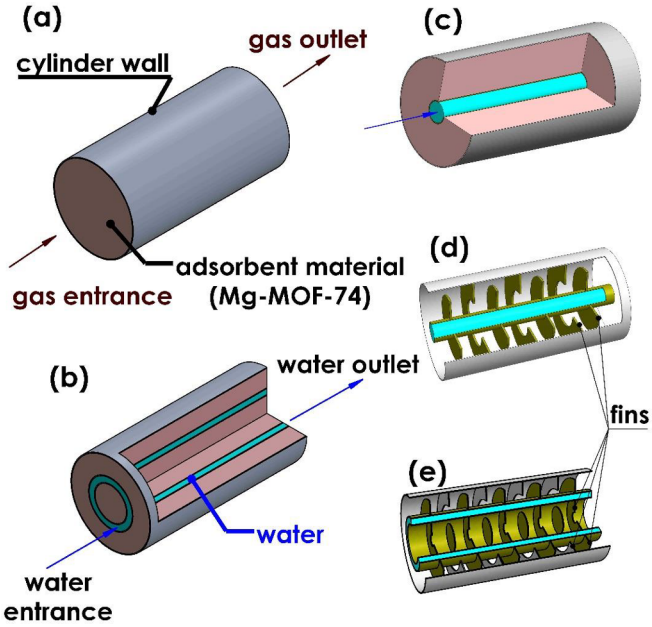


Fig. 2. Adsorption bed configurations, (a) base bed, (b) bed with annular water cooling, (c) bed with inner water cooling, (d) bed with inner water cooling and fins, and (e) bed with annular water cooling and fins.

pipe at fin-pitching of 50, 25, and 12 mm, as shown in Fig. 2d and e. The bed has a radius of 25 mm and a length of 200 mm. Thus, it has an aspect ratio (*AR*) of length to a radius of 8. In this work, the impact of the aspect ratio on the adsorption process is analyzed, so different aspect ratios are examined by changing the pipe length to 100, 150, 300, and 600 mm while keeping the total bed volume the same. Consequently, the bed radius is adjusted to keep this constraint of the volume.

In all investigated cases, the gas mixture enters the adsorption bed at 297 K and 101.3 kPa. In addition, the outer wall of the bed is considered to be exposed to natural-convection during the adsorption process. For the configurations where water flow is introduced, it is assumed that the water enters the bed at 292 K with a flow rate of 1 L/min.

It is worth mentioning that the use of Mg-MOF-74 is an example of excellent adsorbents. However, thermal management investigated in this paper is valid for all types of beds and adsorbents. Mg-MOF-74 was reported to be economically feasible when the flue gas is dehydrated [24]. For a dry N_2 and CO_2, this adsorbent shows excellent performance [1,40]. However, its synthesis process is cumbersome and expensive [41].

3. Governing equations

The flow condition is considered to be laminar since the Reynolds number is small [42]. Hence, the conservation equations that control this type of flow are continuity, momentum, energy, and species transport. The equation of transport species can be written in the form of convection-diffusion equation in which the local mass-fraction of each species is calculated as following [40]:

$$\varepsilon\frac{\partial(\rho y_i)}{\partial t} + \nabla.(\rho\vec{v}y_i) = -\nabla(\varepsilon\rho D_{disp,i}\nabla y_i) - (1-\varepsilon)\rho_P M_i\frac{\partial q_i}{\partial t} \quad (1)$$

where ε is the bed porosity, ρ is the gas mixture density (kg m^{-3}), y_i is the mass fraction of species i (i.e., CO_2, N_2 or He), $\vec{v}$ is the velocity vector, $D_{disp,i}$ is mass coefficient dissipation, M_i is the molecular weight of the species i and q_i is the uptake of species i (mol kg^{-1}). The first and second terms in the above equation represent the unsteady mass and the advection mass transfer while the third and fourth terms represent the mass diffusion and mass source. The adsorption kinetic $(\partial q_i/\partial t)$, which affect the mass source in the continuity equation, can be obtained from the linear driving force (LDF) model, as described below [43,44]:

$$\frac{\partial q_i}{\partial t} = k_{L,i}(q_i^* - q_i) \quad (2)$$

where q_i* is the adsorbed amount of species i at equilibrium state while q_i is the actual amount that adsorbed from the species i, $k_{L,i}$ is adsorption time constant, and it has a value of 0.1182 (s^{-1}) and 0.3043 (s^{-1}) respectively for CO_2 and N_2 species [24,45]. The q_i* can be evaluated from the fitting of the experimental isotherm using Toth model as following [45]:

$$q_i^* = \frac{q_{m,i}K_{eq,i}y_iP}{(1+(K_{eq,i}y_iP)^{ni})^{1/ni}} \quad (3)$$

where $q_{m,i}$ is the maximum adsorbed amount of the species i (mol kg^{-1}), P is the pressure (Pa), ni and $K_{eq,I}$ are adsorption constants where the former is obtained from the experimental fitting while the later can be calculated as a function of temperature as following [45]:

$$K_{eq,i} = k_0 e^{-\left(\frac{\Delta H_i}{RT}\right)} \quad (4)$$

where k_0 is a constant that does not depend on the temperature, ΔH_i is the adsorption heat (J mol^{-1}), and it varies according to the species and sometimes according to the amount of the adsorbed gas itself, R is the universal gas constant (8.314 J mol^{-1} K^{-1}), and T is the gas temperature (K).

The overall mass (continuity) equation is [46]:

$$\varepsilon\frac{\partial(\rho)}{\partial t} + \nabla\cdot(\rho\vec{v}) = -(1-\varepsilon)\rho_p\sum_i M_i\frac{\partial q_i}{\partial t} \quad (5)$$

In addition to the continuity equation (Eq. (1)), the momentum transport equation can be expressed as [46]:

$$\varepsilon\frac{\partial(\rho\vec{v})}{\partial t} + \nabla\cdot(\rho\vec{v}\vec{v}) = -\nabla p + \nabla\cdot\bar{\bar{\tau}} + \rho\vec{g} + S_{momentum} \quad (6)$$

where $\bar{\bar{\tau}}$ is stress tensor, $\vec{g}$ is the gravity acceleration vector, and $S_{momentum}$ is the momentum source term. According to the Ergun equation, the momentum source term can be calculated as [46]:

$$S_{mementum} = -\left(\frac{\mu}{\kappa}v_i + C_2\frac{1}{2}\rho\,|\vec{v}|\,v_i\right) \quad (7)$$

where μ is the gas dynamic viscosity (Pa s), $1/\kappa$ is the viscous resistance of the porous media (m^{-2}) and C_2 is the inertia resistance (m^{-1}).

There are two different zones; porous zone and solid zone, hence the energy equation for each zone is expressed differently. The heat transfer inside the porous media is governed by the energy conservation equation which describes a balance between the stored energy in the adsorbent bed and the energy change due to thermal diffusion, convective flow, and pressure work, in addition to the generated/consumed energy from the process of adsorption/desorption. Accordingly, the energy conservation equation can be expressed as:

$$\frac{\partial}{\partial t}[\varepsilon\rho E_g + (1-\varepsilon)\rho_b E_s + \nabla\cdot(\vec{v}(\rho E + P))] = \nabla\cdot\left(k_{eff}\nabla T - \sum_i h_i\vec{J_i} + \bar{\bar{\tau}}\cdot\vec{v}\right) + (1-\varepsilon)\rho_b\sum_i \Delta H_i\frac{\partial q_i}{\partial t} \quad (8)$$

where E_g and E_s are the total gas energy and total adsorbent energy (J mol^{-1}), ρ_b is the bed density (kg m^{-3}), h_i is the sensible enthalpy (J kg^{-1}) and $\vec{J}$ is the diffusion flux (kg m^{-2} s^{-1}). k_{eff} is the effective thermal conductivity of the adsorbent; it can be calculated as a function of the gas thermal conductivity k_g and the adsorbent thermal conductivity k_s [45]:

$$k_{eff} = \varepsilon k_g + (1-\varepsilon)k_s \quad (9)$$

On the other hand, the heat transfer through the walls and fins (i.e. solid zone) is expressed by the balance between the heat stored in the wall and fins and the diffusion heat through it [45].

$$\frac{\partial}{\partial t}(\rho_w C_w T_w) = \nabla(k_w\nabla T_w) \quad (10)$$

where ρ_w, C_w and k_w are the wall/fin density, specific heat capacity, and thermal conductivity. T_w is the local wall/fin temperature.

Since the outer wall of the bed is exposed to natural convection with ambient air, the heat transfer coefficient may be obtained from the following correlation of Nusselt number [47]:

$$Nu = \frac{h_{ext}D_o}{k_{air}} = \left[0.60 + 0.387Ra^{1/6}\Big/\left[1+\left(\frac{0.559}{\mathrm{Pr}_{air}}\right)^{8/27}\right]\right]^2 \quad (11)$$

where D_o, h_{ext}, and, k_{air} are the external bed diameter, heat transfer coefficient, and thermal conductivity of the air, respectively. Pr_{air} and Ra are the Prandtl number and Rayleigh number, respectively.

4. Numerical procedure

In order to perform the CFD simulations, the ANSYS-Fluent program is used. The code utilizes the finite volume method to discretize the differential equations. The adsorption equation and source terms are defined as a user-defined-function (UDF) written in C and linked to ANSYS Fluent. A SIMPLE scheme is used to couple the pressure-velocity equations for the solution of the transport equations.

Furthermore, it is clear from Fig. 2 that the problem can be solved as 2D axisymmetric [40]; hence, the computational expense would be reduced. Fig. 3 depicts the computational domain of all the investigated configurations. The thermal properties of the solid wall and the adsorbent material are summarized in Tables 1 and 2.

It should be noted that the developed model is valid under all conditions (T and P) that are stable for adsorbent materials. The temperature should be lower than the thermal stability limit of adsorbents, i.e., before melting the linkers of MOFs or organic materials. Likewise, the pressure should not be destructive. A medium pressure of 50 MPa was investigated for Mg-MOF-74 [23].

5. Results and discussion

5.1. Mesh study and model validation

To choose the correct grid size and time-step size during our time-dependent tow-dimensional numerical simulations, a mesh and time-step size dependence study was carried out to make sure the solution does not depend on the mesh size or time step. The study was conducted for a bed of 200 mm in length and 25 mm in diameter. Mesh size that contains 1,500 to 33,500 cells has been tested with time steps of 60, 50, 40, and 30 s. Fig. 4 shows the breakthrough curves for CO_2 and N_2 for different mesh sizes and various time steps. It turns out that the solution becomes independent of the mesh size when 22,000 cells are applied in the computational domain, in which the maximum error is less than 1.5% in the carbon dioxide uptake, and it is about 1% in terms of CO_2 concentration ratio. This number of cells is corresponding to a cell length of 0.5 mm. Furthermore, the solution becomes independent of the time step when a value of less than or equal 40 s is utilized, which gives an error of 2.5% in the CO_2 uptake, while the error in the average bed temperature and the heat transfer rate is less than 1%. Accordingly, a time step of the 30 s is adopted for this study for more accurate results.

The numerical model is validated against experimental data of the authors [24,40,45], where the model showed an excellent agreement in the breakthrough curves of CO_2 and N_2 concentration ratios with those of experimental data. More details about the model validation are explained in the Appendix A.

5.2. Adsorption/desorption dynamics

The geometry configuration of Fig. 2.a is adopted to analyze the adsorption and desorption processes. The system consists of the adsorbent, Mg-MOF74, and it is surrounded by a wall cooled by natural convection of ambient air at 27 °C. The gas mixture enters the adsorbent

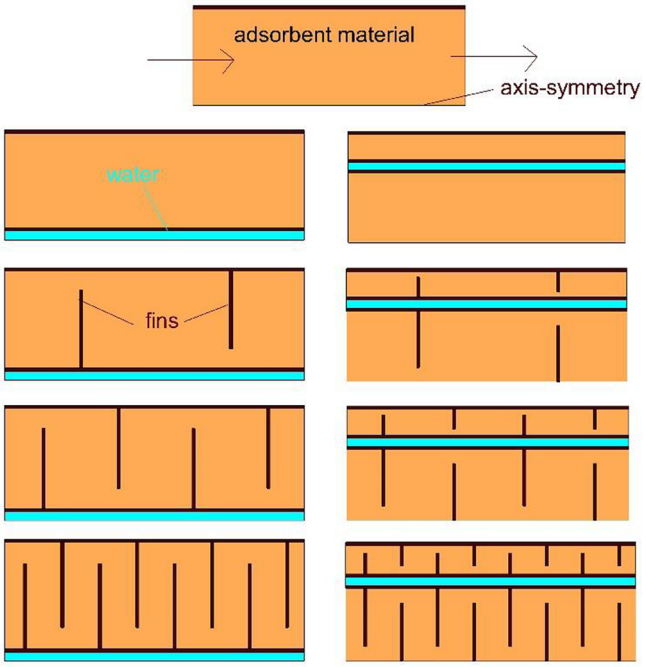


Fig. 3. Computational domains representation.

Table 1
Thermal properties of the solid wall.

Property	Cylinder wall
Material type	Stainless steel
Density (kg/m^3)	7941
Thermal conductivity (W/m K)	15.14
Heat capacity (J/kg K)	358.98 T + 0.487394 T^2-2.65708e-07 T^3

bed at 297 K, 20 wt% CO_2, and 80 wt% N_2. The boundary conditions for this case are summarized in Table 3.

Fig. 5 shows the breakthrough curves of CO_2 and N_2. Until about 14 min of gas mixture feeding, the adsorbent is able to absorb all CO_2 from the mixture. Thus, only nitrogen is observed at the bed outlet. Then, the mole fraction of CO_2 at the bed outlet rises gradually until the concentration ratio between the bed outlet and the initial feeding concentration (C/C_0) reaches almost the unity, which indicates the bed is fully saturated with CO_2. Up to the first 14 min of the adsorption process, the CO_2 uptake remains almost constant by adsorbing 0.108 mol/kg at each time step of the calculation, so it increases linearly during this period as shown in Fig. 6. This indicates that the adsorbent material gets saturated by carbon dioxide, and it is no longer able to adsorbs more CO_2; hence, all the carbon dioxide fed to the bed leaves it.

Additional details on the adsorption process are demonstrated in Fig. 7 that shows the contours distribution of the CO_2 uptakes. History of CO_2 uptakes profiles at three points, i.e., the bed inlet, middle, and outlet, is shown in Fig. 7b. As the gas mixture enters the adsorbent bed, the Mg-MOF74 adsorbs CO_2 gas where most of the adsorption process takes place near the entrance. After a while, the adsorbent near the entrance reaches saturated with the CO_2. This can be noticed from Fig. 7b by looking at the curve of point "A" which located at the entrance of the bed. The curve becomes horizontal when the adsorbent material reaches saturated. The point 'B' starts adsorbing CO_2 after about 250 s from the beginning of gas mixture feeding. The CO_2 uptake value increases rapidly initially, and then after a short period, the increases become gradually until the adsorbent at that position, point "B," becomes fully saturated with CO_2.

The adsorption process is known to be an exothermic reaction. Consequently, the local temperature increase (Fig. 8), leading to a reduction in the local CO_2 adsorption. Again, by monitoring the three points of bed entrance (point A), bed central (point B) and bed outlet (point C), one can notice that at the entrance, the temperature rises to a peak value of 315 K, while the bed central and the bed outlet temperatures rise to around 360 K when they start to adsorb CO_2 gas. The reason behind that could rely on the relatively cold mixture

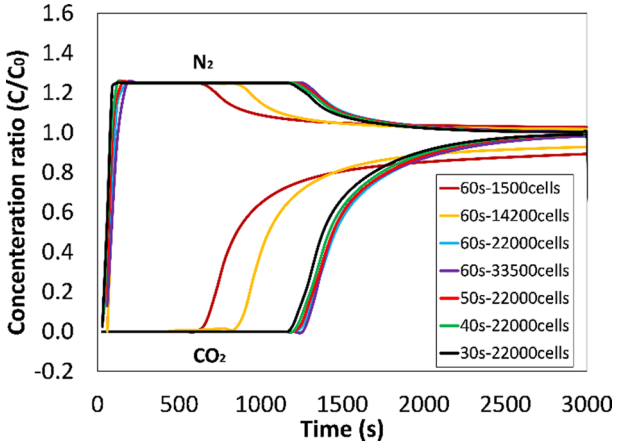


Fig. 4. Breakthrough curves for CO_2/N_2 at different mesh sizes and different time steps.

Table 3
Boundary conditions for the base case.

Parameter	Value
Velocity inlet (m/s)	0.02
Inlet temperature (K)	297
Pressure (Pa)	101,325
Ambient temperature (°C)	27
CO_2 inlet molar fraction	20%
N_2 inlet molar fraction	80%

temperature at the entrance (292 K), which cools the bed; therefore, point 'A' has a small peak temperature compared to the other points 'B' and 'C'. The small peak temperature value of point 'A' demonstrates the high uptake values at that point (~6 mol/kg) as shown in Fig. 7b. On the other hand, point 'B' and 'C' are located far away from the entrance, thus, the mixture gas reaches these points at a relatively high temperature owing to the exothermic reaction, and it consequently limits the CO_2 uptake. Therefore, the maximum CO_2 uptake reduces as we moved away from the entrance. The maximum CO_2 uptakes at points A, B, and C are 5.95, 4.75, and 4.10 mol/kg, respectively.

To analyze the desorption of CO_2 on the bed, the feeding gas mixture is stopped, and the cylinder outlet is closed while the gas entrance is opened to allow the adsorbed CO_2 to leave the bed. The bed is heated up to 370 K at the outer circumference wall, so as the adsorbent temperature rises, it releases the adsorbed carbon dioxide. Fig. 9 shows an adsorption and desorption process for an adsorbent bed with a radius

Table 2
Thermal properties of the adsorbent material [40].

Property	Adsorbent
Material type	Mg-MOF74
Density (kg/m^3)	230.1
Thermal conductivity (W/m K)	0.3
Heat capacity (J/kg K)	900
Particle density, ρ_p (kg/m^3)	911
Bed porosity, ε	0.7417
Max CO_2 adsorption concentration, $q_{m,co2}$ (mol/kg)	11.4048
Max N_2 adsorption concentration, $q_{m,N2}$ (mol/kg)	6.7072
CO_2 adsorption constant, $K_{o,CO2}$ (Pa^{-1})	3.089×10^{-11}
N_2 adsorption constant, $K_{o,N2}$ (Pa^{-1})	9.36×10^{-10}
CO_2 heat of formation (Eq. (4)), ΔH_{CO2} (J/mol)	−42,000
N_2 heat of formation (Eq. (4)), ΔH_{N2} (J/mol)	−18,000
CO_2 mass transfer coefficient, $K_{L,CO2}$ (s^{-1})	0.1182
N_2 mass transfer coefficient, $K_{L,N2}$ (s^{-1})	0.3043
CO_2 heat of adsorption (Eq. (8)), ΔH_{CO2} (J/mol)	$-(42492.6 - 6568.83q + 3973.75q^2 - 959.838q^3 + 69.1208q^4)$ J mol^{-1} ($0 < q < 7.5$ mmol g^{-1})
N_2 heat of adsorption (Eq. (8)), ΔH_{N2} (J/mol)	−18,000

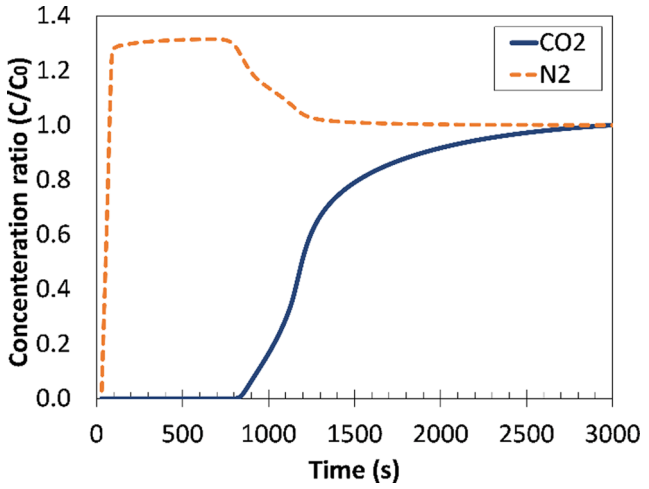


Fig. 5. The breakthrough curves for N_2/CO_2 during the adsorption process at CO_2/N_2 inlet of 20/80 (% by volume), 297 K, 101.3 kPa.

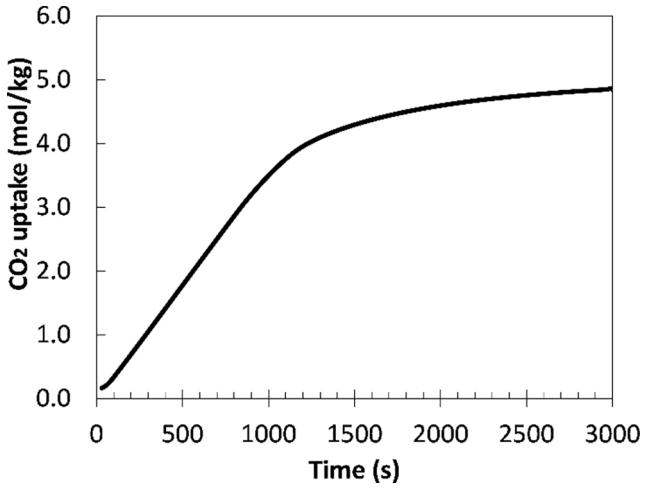


Fig. 6. Carbon dioxide uptakes (as the average value of the entire bed) during the adsorption process (inlet conditions: P = 101.325 kPa and T = 297 K).

and length of 25 and 200 mm, respectively. Due to the low value of the thermal conductivity of Mg-MOF74 (0.3 W m^{-2} K^{-1} [1]), and hence low thermal diffusivity, it takes a longer time for the bed to release the adsorbed amount of CO_2 from 4.85 to 3.27 mol/kg.

5.3. Effect of aspect ratio (AR)

The above analysis was conducted for the adsorbent bed with a 25 mm radius and 200 mm length (i.e. $AR = L/R = 8$). In this section, the effect of aspect ratio on the adsorption and desorption processes is analyzed by varying the aspect ratio from 2.8 to 41.6. This is achieved by changing the bed length to be 100, 150, 200, 300, and 600 mm and adjusting the cylinder-bed radius in such a way the bed volume remains constant for all cases. Therefore, the amount of adsorbent material is similar in all cases.

Fig. 10 shows the carbon dioxide uptakes during the adsorption/desorption processes for the adsorbent bed, having different aspect ratios. It turns out that increasing the aspect ratio from 2.8 to 41.6 leads to an improvement in the CO_2 uptakes from 4.46 to 5.03 mol/kg (13% improvement) when the adsorption process is allowed to take place until the CO_2 concentration reaches 90% of the initial CO_2 concentration ($C/C_o = 90\%$). This improvement in the CO_2 uptake is attributed to many parameters. In fact, the increase of the aspect ratio from 2.8 up to 41.6 increases the cylinder surface area by about 2.5 times, which

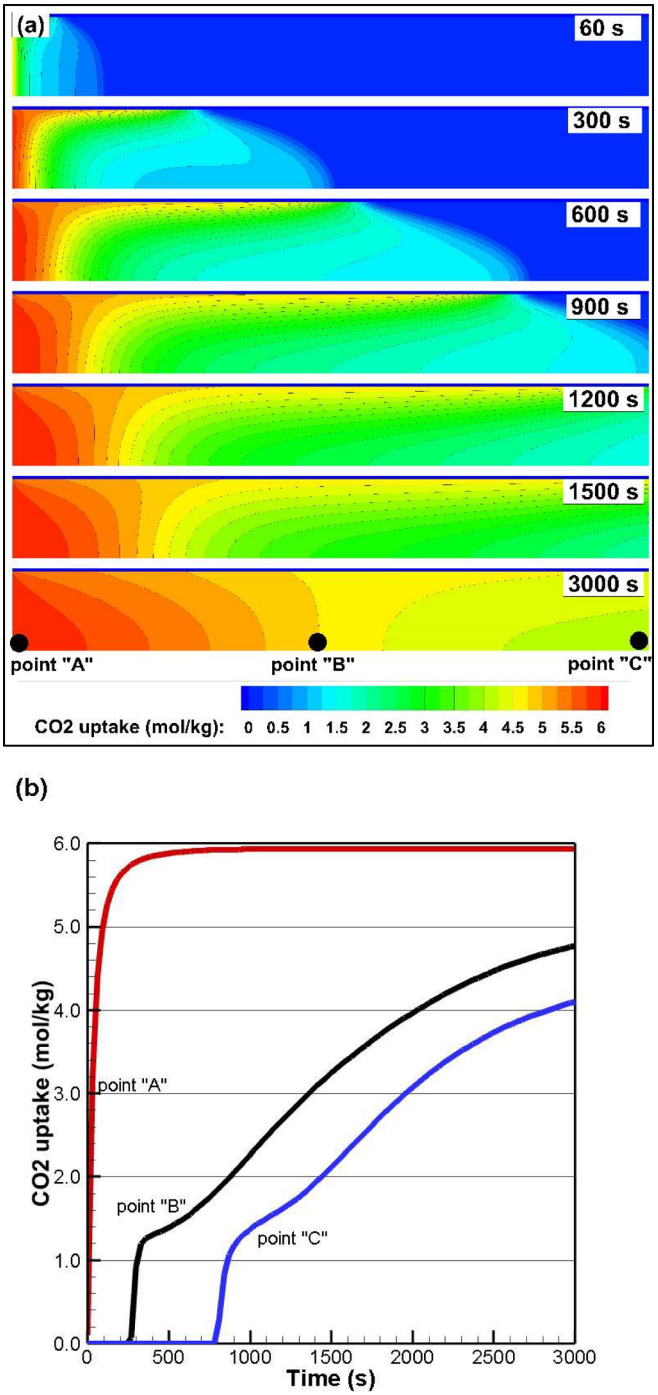


Fig. 7. CO_2 uptake in a format of a) contours and b) history profile at three points (bed entrance, middle, exit) (inlet conditions: P = 101.325 kPa and T = 297 K).

enhances the heat dissipation of the generated heat due to the exothermic adsorption process. The high aspect ratio bed has a small cylinder radius; not only the adsorbent material located near the bed-wall would have a chance to dissipate further heat to the surroundings, but even the adsorbent bulk located at the bed middle would have a reasonable heat dissipation rates due to small bed thickness. Moreover, the forced convection cooling of the feeding gas plays a significant rule in reducing the bed temperature, which directly influences the amount of CO_2 uptakes. The feeding gas enters the bed at 297 K with a mass flow rate of 5×10^{-5} kg/s. Therefore, in the cases where a large aspect ratio is utilized (e.g., $AR = 41.6$), the flow moves with a high-velocity value

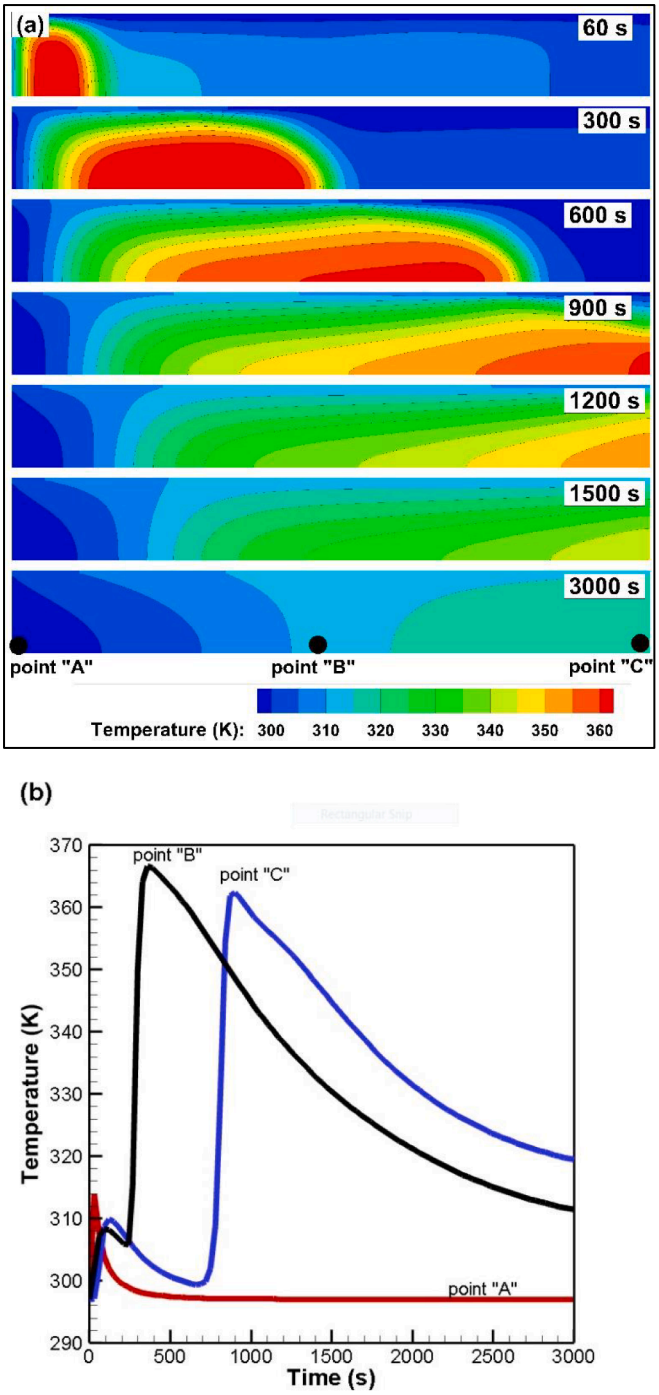


Fig. 8. Temperature distribution in a format of a) contours and b) history profiles at three points (inlet conditions: P = 101.325 kPa and T = 297 K).

because the mass flow rate is kept similar in all cases while the cross-section area changes. Hence, higher forced convection cooling is obtained in the case of *AR = 41.6* than in the other cases of smaller aspect ratio. Accordingly, the average bed temperature decreases dramatically (at high *AR*), as shown in Fig. 11, which depicts the average bed temperature during the adsorption/desorption process.

On the other hand, the feeding gas enters the bed of a large aspect ratio at a small velocity value, so the effect of forced convection cooling becomes weak. Consequently, the temperature increases dramatically, especially for a thicker bed. Fig. 12 represents the temperature contours for different aspect ratios after 300 s of the beginning of the adsorption

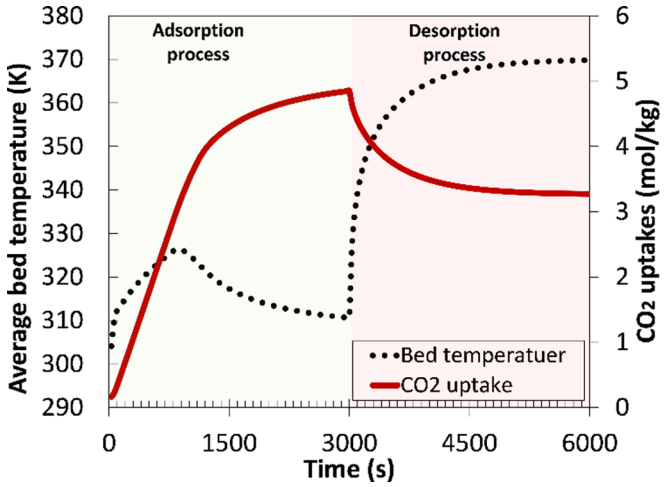


Fig. 9. Adsorption and desorption processes in a cylindrical bed of (R = 25 mm and L = 200 mm, adsorption: 20/80 (CO_2/N_2 % vol.), T = 297 K, P = 101.3 kPa, and desorption: outer wall temperature = 370 K).

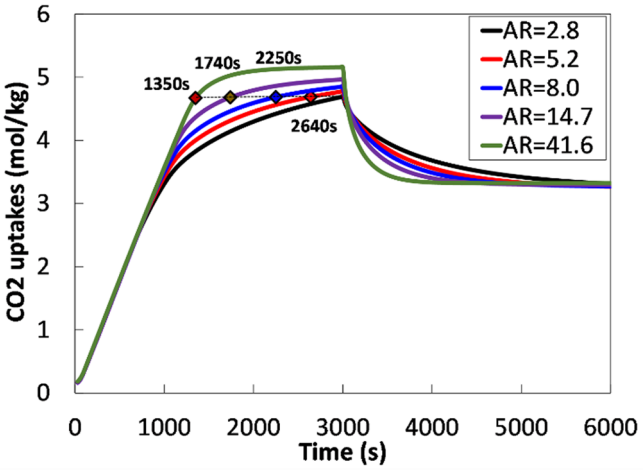


Fig. 10. Comparison of the adsorption/desorption process at different aspect ratios (inlet conditions: P = 101.325 kPa and T = 297 K).

process.

The adsorption process is performed for 50 min, and then the

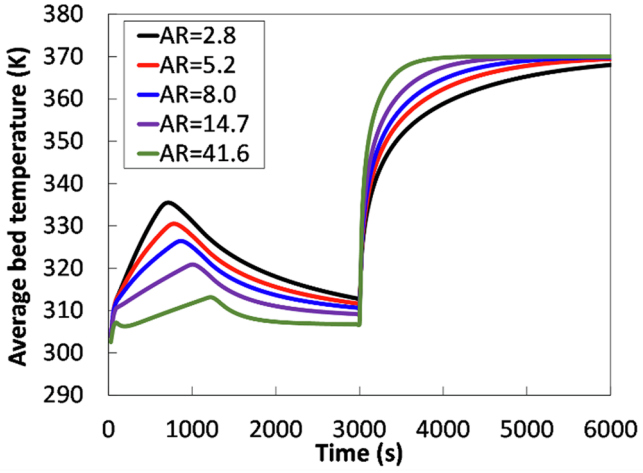


Fig. 11. Average bed temperature during adsorption/desorption processes (P = 101.325 kPa and T = 297 K).

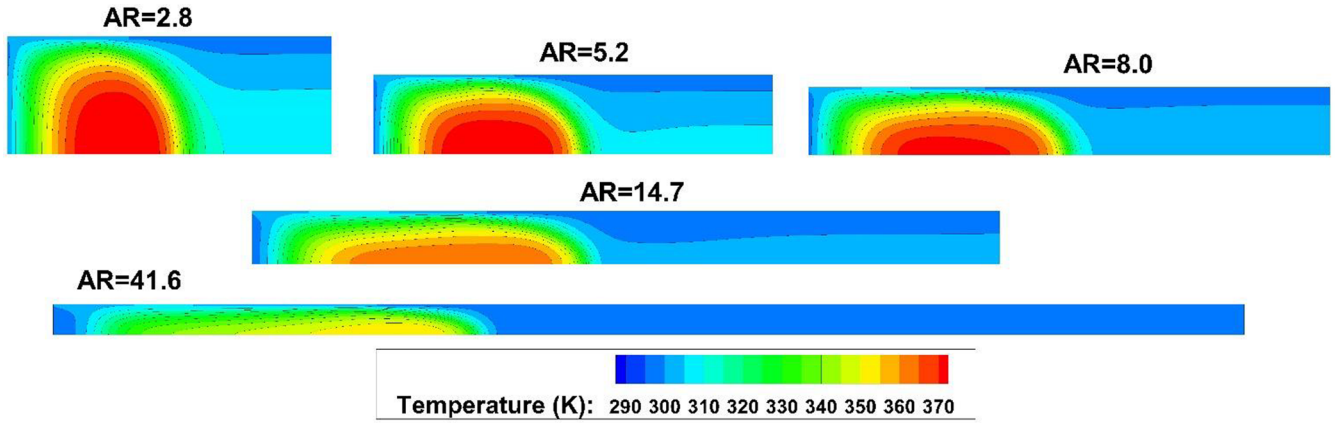


Fig. 12. Temperature contours after 300 s of the adsorption process (inlet conditions: P = 101.325 kPa and T = 297 K).

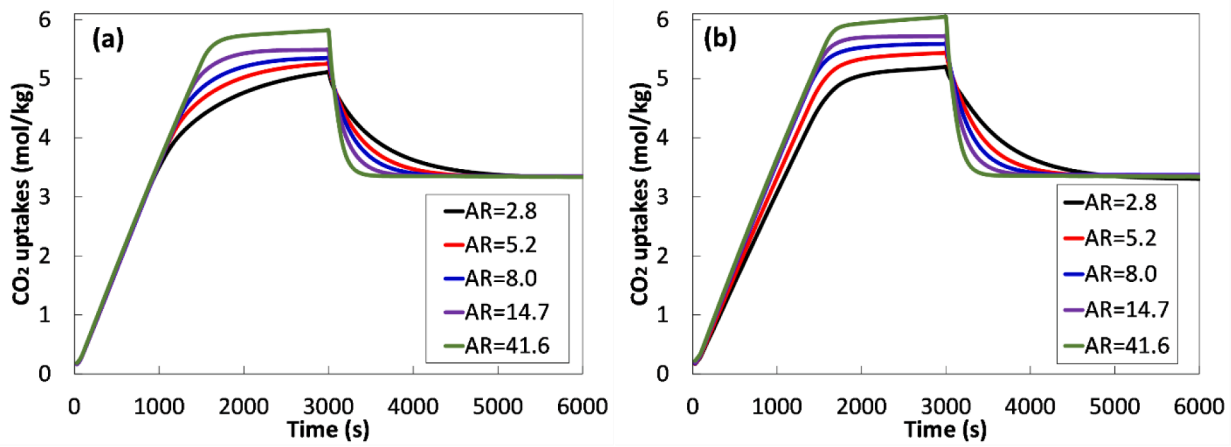


Fig. 13. Comparison of adsorption/desorption process for cases cooled by a) water pipe only b) water pipe with fins (Adsorption at P = 101.325 kPa, T = 297 K and T_{water} = 292 K. Desorption at P = 101.325 kPa, T_{wall} = 370 K, and T_{water} = 370 K).

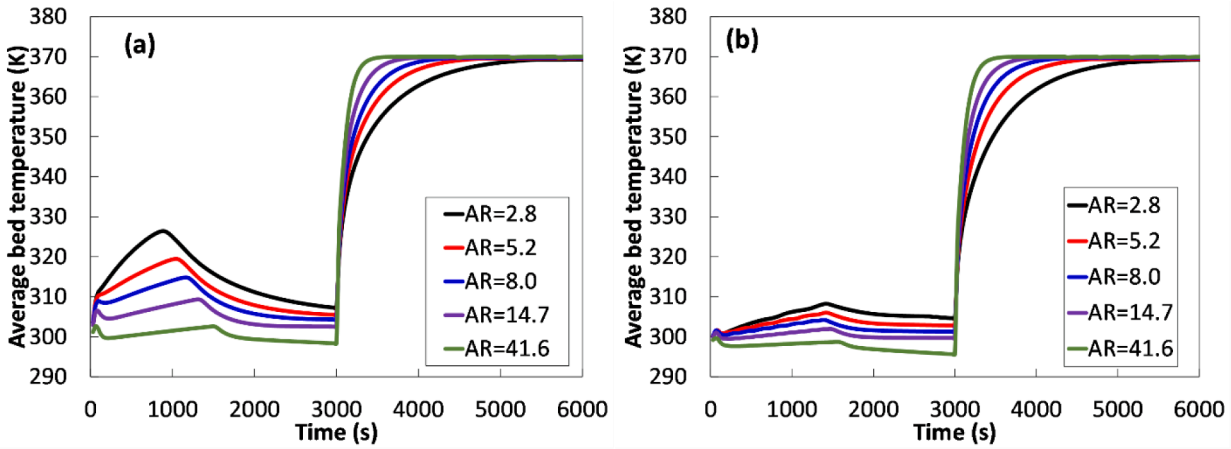


Fig. 14. Average bed temperature for cases cooled by a) water pipe only b) water pipe with fins (Adsorption at P = 101.325 kPa, T = 297 K and Twater = 292 K. Desorption at P = 101.325 kPa, Twall = 370 K, and Twater = 370 K).

desorption process takes place for another 50 min. Again, the higher aspect ratio cases show a better regeneration (desorption) process than the smaller aspect ratio cases (see Fig. 10). It takes 810 s for the case of *AR* = 41.6 to reduce CO_2 uptakes from 5.16 to 3.35 mol/kg while the case of *AR* = 2.8 takes 2550 s to reduce CO_2 uptakes from 4.69 to 3.35 mol/kg. Thus, the cyclic capacity is only 1.81 mol/kg for *AR* = 41.6 and 1.34 mol/kg for *AR* = 2.8. Of course, raising the bed wall temperature to higher values improves the desorption process by dropping the CO_2 uptakes limit to a smaller value less than 3.35 mol/kg. This limitation in the residual CO_2 uptake is restricted by the heating temperature (370 K). Hence, raising the heating temperature to a value greater than 370 K would reduce the residual CO_2 on the bed to less than 3.35 mol/kg.

Also, from Fig. 10, one can notice that the case of *AR* = 2.8 adsorbs up to 4.69 mol/kg in 3000 s. In comparison with other cases, this adsorption capacity (4.69 mol/kg) is achieved in a shorter time as the

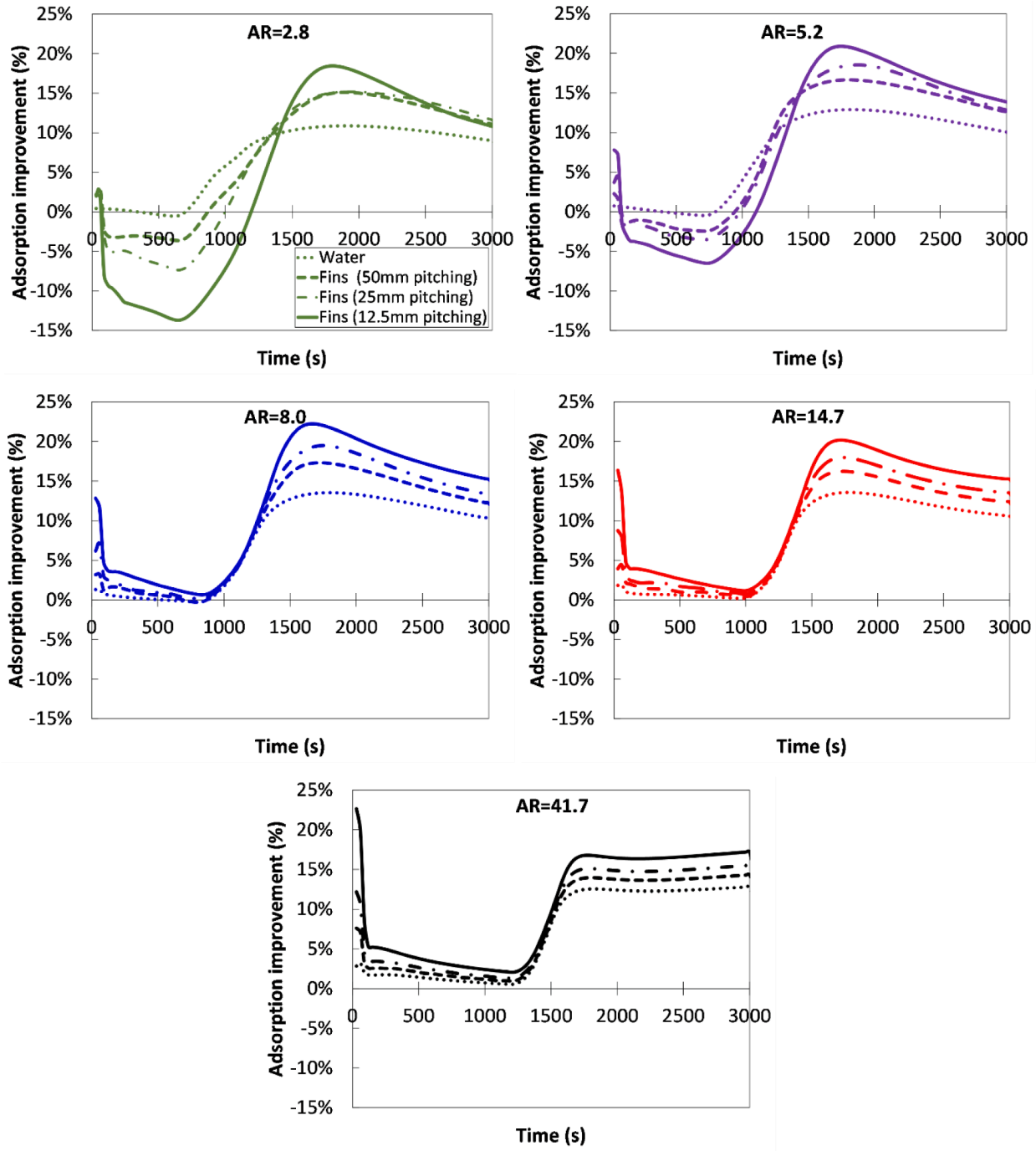


Fig. 15. Improvement percentage in CO_2 uptake due to applying cooling water in the central pipe (inlet conditions: P = 101.325 kPa, T = 297 K, and T_{water} = 292 K).

aspect ratio increases. If a horizontal line is drawn for a CO_2 adsorption value of 4.69 mol/kg as in Fig. 10, it can be observed that the case of *AR* = 41.6 adsorbs the same amount of CO_2 uptake as in the case of *AR* = *2.8* for but in small period (1350 s), which implies a time-saving by double. A similar observation is also noticed in the desorption process where it takes the desorption process takes 2550, 1860, 1560, 1260, and 810 s to reduce CO_2 uptakes to 3.35 mol/kg for the cases of AR = 2.8, 5.2, 8.0, 14.7, and 41.6 respectively.

5.4. Cooling effects on the adsorption process

This section examines the influence of cooling and heating the adsorbent bed through different geometrical configurations associated with cooling/heating water conduit and bed fins. Eight cooling/heating water configurations are tested and compared to a base case (BC) of conventional bed cooled by natural convection of ambient air and heated by a heater jacket at the outer surface. These configurations are explained in Fig. 3. The eight configurations are classified into two categories; the first one is cooled by introducing a pipe in the central bed axis. Three cases of this category have fins attached to the pipe (fin-pitch is 50, 25, and 12.5 mm). The second category is cooled by an annular pipe located at a distance that divides the adsorbent bed volume equally. Also, three cases of this category have fins similar to the first category.

Figs. 13 and 14 display CO_2 uptakes and average bed temperature during the adsorption/desorption processes at different aspect ratios of

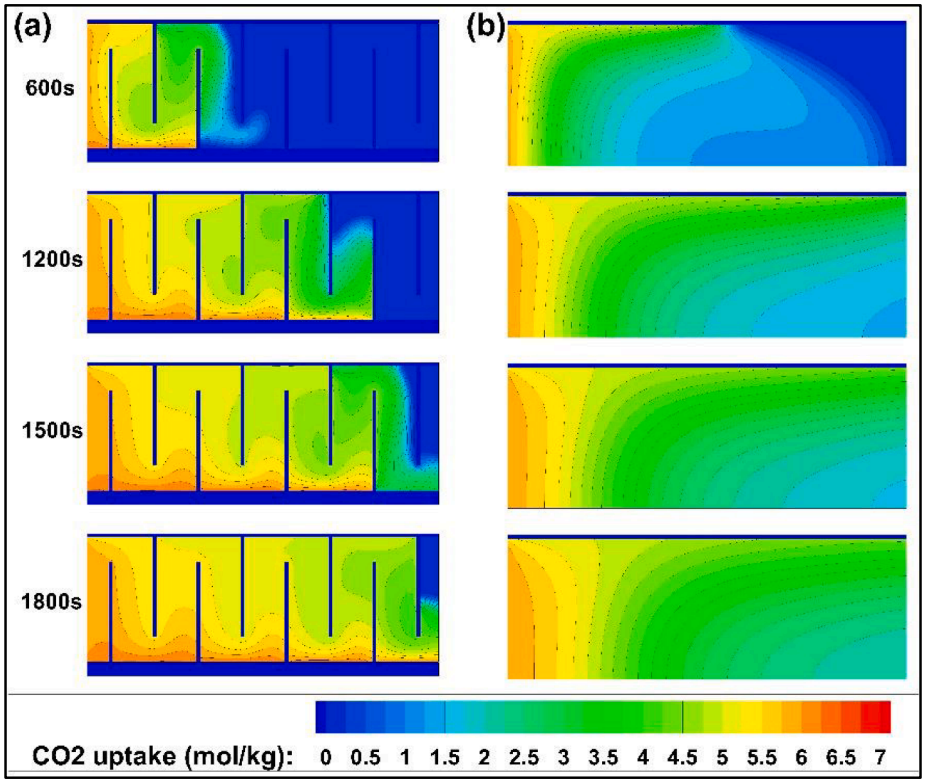


Fig. 16. CO_2 uptakes comparison a) finned-pipe (AR = 2.8, fin-pitch = 12.5) b) base case (AR = 2.8) (inlet conditions: P = 101.325 kPa, T = 297 K, and T_{water} = 292 K).

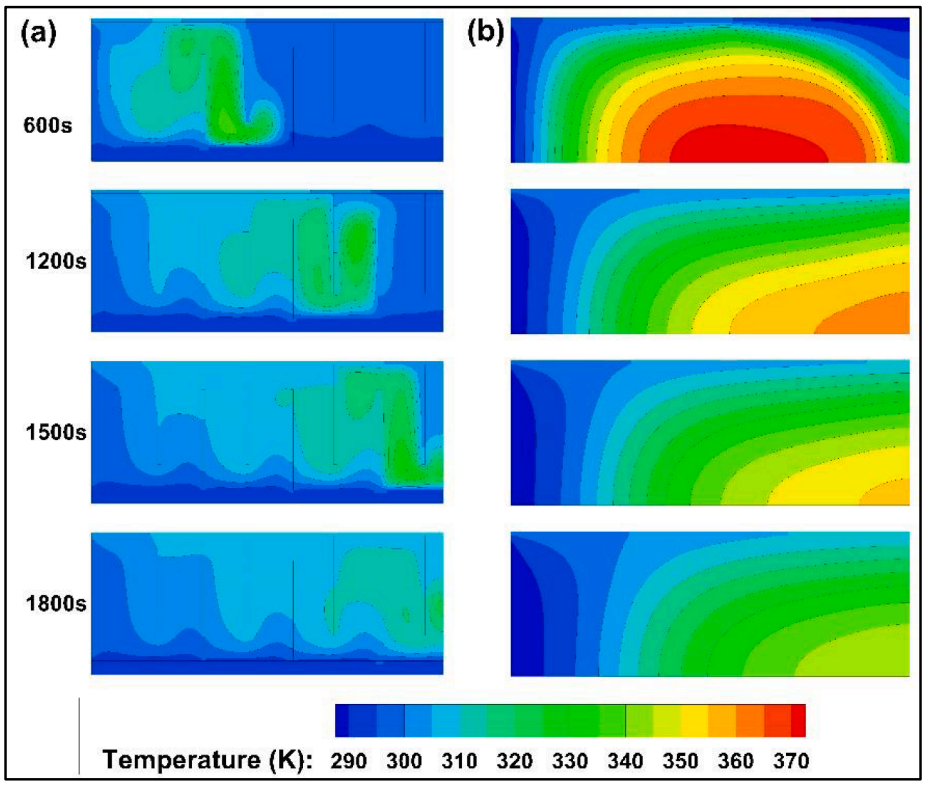


Fig. 17. Temperature contour comparison a) finned-pipe (AR = 2.8, fin-pitch = 12.5) b) base case (AR = 2.8) (inlet conditions: P = 101.325 kPa, T = 297 K and T_{water} = 292 K).

the adsorbent bed that cooled by either water pipe only or water pipe attached by fins (12.5 mm pitch). Cooling the bed with water pipe slows the temperature growths in the bed because it quickly dissipates the generated from the CO_2 adsorption process; hence, more CO_2 uptakes are achieved. Further reduction in the average bed temperature is attained by implementing fins to the pipe (see Fig. 14b**)**. Whereas, for the

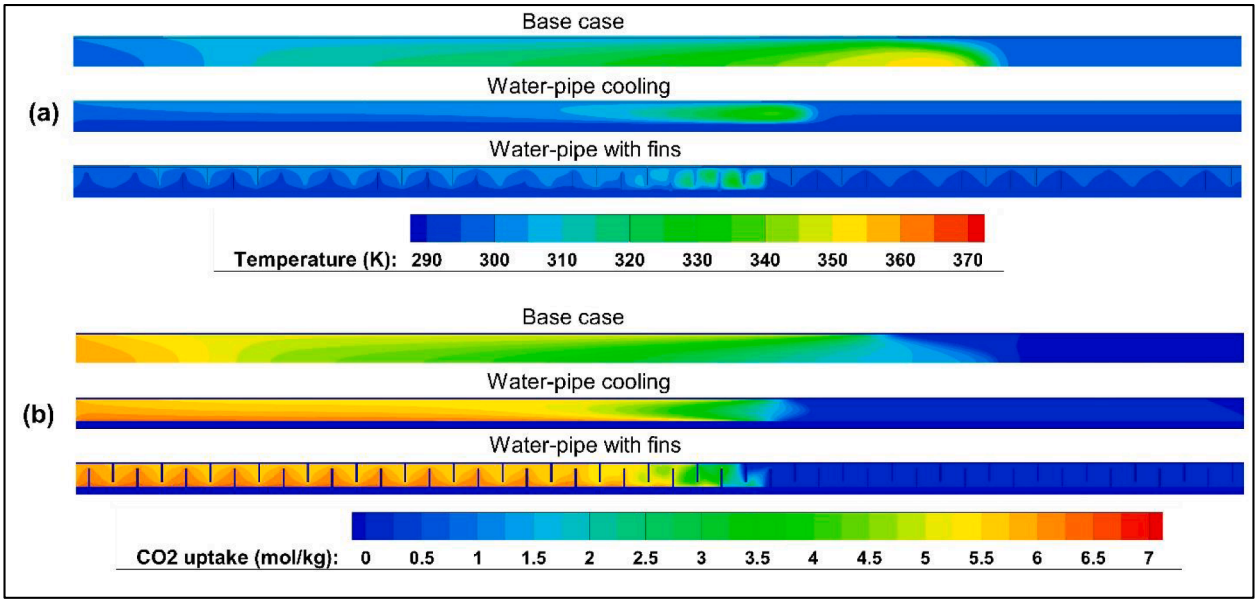


Fig. 18. Contours of a) Temperature and b) CO_2 uptakes for AR = 41.6 at 900 s (inlet conditions: P = 101.325 kPa, T = 297 K, and T_{water} = 292 K).

case with *AR* = 2.8, the peak bed temperature reduces from 326.4 K to 308.3 K when fins are attached to the water pipe. Besides, the time at which peak bed temperature occurs is delayed, which indicates that the bed can adsorb more CO_2 before its temperature escalates.

Fig. 15 summarizes the improvement profiles at the bed center for the cases cooled by a water pipe and the base case. The maximum improvement in the CO_2 uptakes can reach up to 10.9–13.6% using only the water pipe while this percentage reaches up to 16.8–22.2% when fins are attached to the pipe with fin-pitch of 12.5 mm. By taking the case of *AR* = 2.8 as an example, there is a degradation in the carbon dioxide uptakes at the beginning of the adsorption process, especially for the case of finned pipe of 12.5 mm fin-pitch. Nevertheless, the CO_2 uptake starts to increase after 1200 s to reach a maximum improvement of 18.5%. This reduction in the CO_2 uptake can be explained by the help of Figs. 16 and 17 that shows the contours of CO_2 uptake and bed temperature at different times, respectively. As the fins obstruct the gas flow, the adsorption process covers less than half of the bed volume at 600 s, while it covers almost the full bed volume in the base case. However, for the bed with finned-pipe, the local CO_2 uptake is larger than in the base case. Then, after 1200 s, the finned-bed adsorbs a similar amount of CO_2 as in the base case when the adsorption process covers only three-quarters of the bed volume.

In contrast, the adsorption process in the base case covers the full bed volume. However, due to the high generated temperature in the base case, the bed adsorbs less amount of CO_2, and it appears in the outlet with a concentration higher than half that of the inlet. So, the finned-pipe lowers the bed temperature (see Fig. 17), which results in higher local CO_2 uptake along the bed. While in the base, higher local uptakes are observed only at the bed entrance (see Fig. 16). Similar observations are also noticed in the larger aspect ratio cases. So, even for small bed diameter (i.e., large aspect ratio), using finnd-pipe with cooling-water would lead to further improvement in the CO_2 uptakes as it results in dissipating the exothermic heat generation coming from the adsorption process. Hence, an increase in the local CO_2 uptake is observed near the water-pipe wall and the fins, as shown in Fig. 18. Table 4 summarizes the CO_2 uptakes and its percentage improvement as well as the time duration for the adsorption process for different aspect ratio and different cooling types (i.e., natural convection is the base case, circular pipe, circular pipe with fins at different pitching distance, annular pipe, and annular pipe with fins at different pitching distance). These quantities are considered when the carbon dioxide concentration reaches 90% of the initial feed concentration (C/C_0 = 0.9), which means the adsorbent bed is no longer adsorbing CO_2, so most of the fed gas passes through the bed without being adsorbed.

Table 4 illustrates that the higher aspect ratio results in higher adsorption quantity and in a short time. On the other hand, if the desorption process is limited to the residual adsorption quantity of 3.35 mol/kg (which is almost the minimum value can be obtained at a heating temperature of 370 K), the higher aspect ratio cases desorb CO_2 reaching the residual adsorption limit in a short time. Accordingly, increasing the aspect ratio results in higher adsorption quantity difference and reduce the duration time of the desorption process dramatically so the adsorption bed would be ready for the next cycle. The water cooling increases the adsorption quantity difference because it increases the CO_2 uptakes during the adsorption process, and since during the desorption process, all cases reach a similar value of residual adsorption quantity (3.35 mol/kg). It indicates that increasing the adsorption quantity during the adsorption process increases the actual operating capacity (adsorption quantity difference).

Furthermore, tremendous improvement is achieved when an annular cooling pipe is introduced to the adsorbent bed, especially at a small aspect ratio, where the improvement percentage recorded to be 35.2%. This percentage jumps to 41.8% when fins attach the annular pipe at 12.5 mm fin-pitch (see Fig. 19a). The annular pipe offers more surface area than that of using a circular pipe at the bed center, and it divided the adsorbent bed to equal volumes. Thus, it cools the adsorbent from two directions, i.e., the inner and outer surfaces of the annular pipe. Accordingly, it dissipates the exothermic heat more efficiently, resulting in enhancing the CO_2 uptake. Fig. 20 shows the temperature and CO_2 uptakes for a bed of *AR* = 2.8 that is cooled by finned-annular-pipe. For lower *AR*, the spacing between the annular pipe and the cylinder wall is large; hence, adding fins leads to more enhancement in the CO_2 uptakes. Contrary, for larger *AR*, where the spacing is minimal, and the annular-pipe water is sufficient to dissipate the heat without needing further fins, all the finned cases improve the CO_2 uptakes by almost 23.5% (see Fig. 19b).

5.5. Pressure losses

The pressure losses through the bed are also calculated for all the investigated cases, as depicted in Fig. 21. It can be noticed from the figure that the pressure losses exceed 1 kPa in all cases of annular pipes having fins (with 12.5 mm fin-pitch) along with the circular pipe. Also, it can be noticed that the increase in aspect ratio from 2.8 to 41.7 leads to a remarkable increase in pressure losses. For example, in the case of an

Table 4
Performance summary of all the investigated cases (P = 101.325 kPa, T = 297 K, and T_{water} = 292 K).

Aspect ratio	Cooling type	CO_2 uptake (@$C/C_{0,CO2}$ = 0.9) (mol/kg)	Improvement of CO_2 uptake (@$C/C_{0,CO2}$ = 0.9)	Time required to reach $C/C_{0,CO2}$ = 0.9 (s)
2.8	Base case	4.46	–	2340
	Circular pipe	4.90	10.6%	2280
	Circular pipe + fins (pitching 50 mm)	5.02	14.6%	2160
	Circular pipe + fins (pitching 25 mm)	5.08	14.7%	2280
	Circular pipe + fins (pitching 12.5 mm)	5.06	17.6%	2010
	Annular pipe	5.82	33.8%	2100
	Annular pipe + fins (pitching 50 mm)	5.85	36.5%	1980
	Annular pipe + fins (pitching 25 mm)	5.89	38.5%	1920
	Annular pipe + fins (pitching 12.5 mm)	5.96	40.8%	1890
5.2	Base case	4.51	–	2130
	Circular pipe	5.06	12.7%	2070
	Circular pipe + fins (pitching 50 mm)	5.20	16.3%	2040
	Circular pipe + fins (pitching 25 mm)	5.29	18.2%	2040
	Circular pipe + fins (pitching 12.5 mm)	5.31	20.2%	1920
	Annular pipe	5.97	33.4%	2040
	Annular pipe + fins (pitching 50 mm)	6.02	35.4%	1980
	Annular pipe + fins (pitching 25 mm)	6.00	36.4%	1890
	Annular pipe + fins (pitching 12.5 mm)	6.05	38.1%	1860
8	Base case	4.60	–	2010
	Circular pipe	5.20	13.3%	1980
	Circular pipe + fins (pitching 50 mm)	5.31	17.0%	1890
	Circular pipe + fins (pitching 25 mm)	5.41	19.1%	1890
	Circular pipe + fins (pitching 12.5 mm)	5.47	21.8%	1800
	Annular pipe	6.04	31.8%	1980
	Annular pipe + fins (pitching 50 mm)	6.05	32.8%	1920
	Annular pipe + fins (pitching 25 mm)	6.07	33.7%	1890
	Annular pipe + fins (pitching 12.5 mm)	6.12	35.1%	1860
14.7	Base case	4.75	–	1890
	Circular pipe	5.39	13.5%	1890
	Circular pipe + fins (pitching 50 mm)	5.49	16.1%	1830
	Circular pipe + fins (pitching 25 mm)	5.57	17.8%	1830
	Circular pipe + fins (pitching 12.5 mm)	5.67	19.9%	1830
	Annular pipe	6.14	28.6%	1950
	Annular pipe + fins (pitching 50 mm)	6.15	29.1%	1920
	Annular pipe + fins (pitching 25 mm)	6.16	29.6%	1890
	Annular pipe + fins (pitching 12.5 mm)	6.17	30.6%	1830
41.6	Base case	5.03	–	1740
	Circular pipe	5.69	12.6%	1830
	Circular pipe + fins (pitching 50 mm)	5.76	14.0%	1800
	Circular pipe + fins (pitching 25 mm)	5.81	15.1%	1800
	Circular pipe + fins (pitching 12.5 mm)	5.90	16.8%	1800
	Annular pipe	6.25	22.9%	1920
	Annular pipe + fins (pitching 50 mm)	6.25	23.2%	1890
	Annular pipe + fins (pitching 25 mm)	6.26	23.4%	1860
	Annular pipe + fins (pitching 12.5 mm)	6.27	23.9%	1830

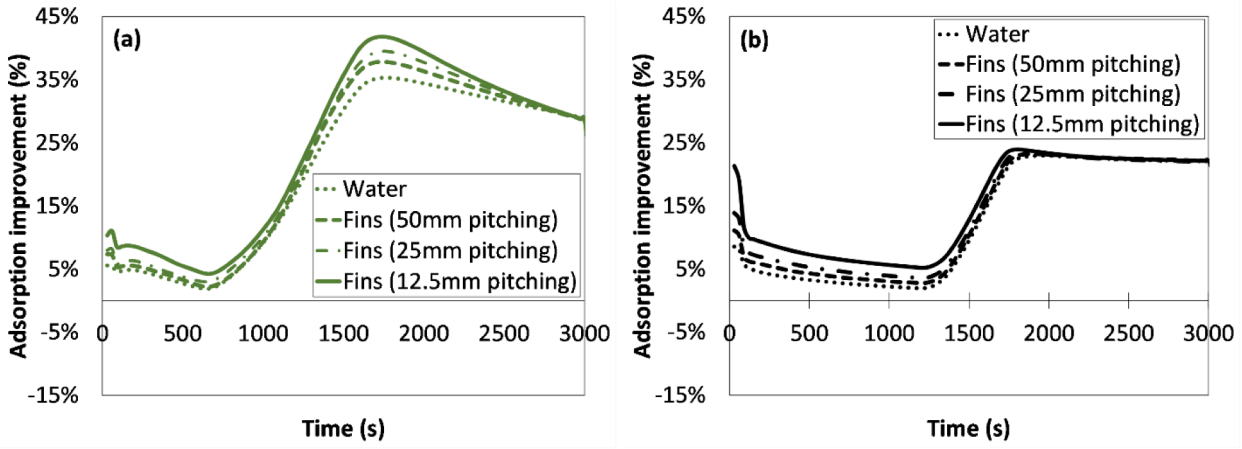


Fig. 19. Improvement percentage in CO_2 uptake due to applying cooling water in an annular pipe at two aspect ratios a) $AR = 2.8$b) $AR = 41.7$ (inlet conditions: P = 101.325 kPa, T = 297 K and, T_{water} = 292 K).

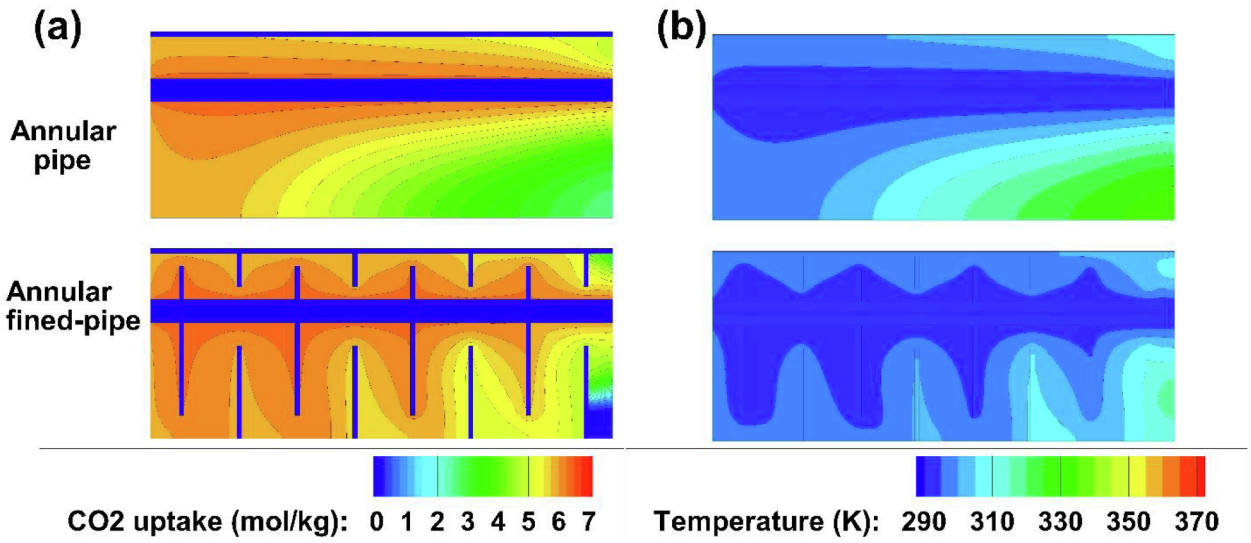


Fig. 20. Contours of (a) CO_2 uptakes and (b) temperature for the annular cooling pipe at 1800 s (AR = 2.8) (inlet conditions: P = 101.325 kPa, T = 297 K, and T_{water} = 292 K).

adsorbent bed with a circular pipe and fins of 12.5 mm fin-pitch, the pressure loss jumps from 102 Pa to 1458 Pa (14 times). Reducing the aspect ratio while keeping the same amount of mass flow rate results in higher velocity value for higher aspect ratio bed, which results in a tremendous pressure loss. In addition, the presence of fins increases the flow path resulting in additional pressure losses; so that, as the number of fins increases (by reducing fin-pitching), the flow path increases leading to amplifying the pressure losses. Accordingly, at a given value of aspect ratio, the pressure loss is intensified as the number of fins is increased. For instance, for an aspect ratio of 41.6, the pressure drop increases gradually from 411 Pa (for the case of circular pipe without fins) to 472, 930, and 1458 Pa for adding fins to the pipe with fin-pitch of 50, 25, and 12.5 mm, respectively. In addition, implementing a circular or annular cooling pipe without any fins results in almost the same value of pressure drop as in the base-case.

6. Conclusions

Enhancing the cooling and heating during adsorption and desorption processes, respectively, are investigated by adequately understanding the correct dynamics of an adsorption bed and using affordable and readily available thermal design and management tools. Adsorbent bed aspect ratio, water cooling, and fins are used to enhance the CO_2 uptake by decreasing the bed temperature during the adsorption process while increasing it during the desorption process. A computational fluid dynamic model is implemented using a user-defined function (UDF, written in C) linked to the ANSYS Fluent program. Mg-MOF-74 is selected as the adsorbent. Different bed aspect ratios (length/radius) of 2.8, 5.2, 8.0, 14.7, and 41.6 are studied to improve the CO_2 working capacity. For all investigated cases, the adsorbent bed volume is kept the same when the aspect ratio changed; thus, they have an equal amount of the Mg-MOF-74. The results reveal that an adsorption bed with a higher aspect ratio (41.6) adsorbs a similar amount of CO_2 (4.46 mol/kg) that adsorbed by the case of $AR = 2.8$, but results in a time saving by half due to minimizing the desorption time. For a specified time of the adsorption process, cases with higher aspect ratio adsorb more CO_2 than those with smaller aspect ratios. Also, it was shown that cooling the adsorbent bed by water passing through a pipe in the center axis of the bed leads to a maximum improvement of 10.9–13.6% depending on the bed aspect ratio.

Furthermore, a remarkable enhancement of 35.2% in the CO_2 uptakes is achieved by using an annular cooling pipe with a bed of small aspect ratio of 2.8. In addition, it turns out that adding fins always improves the CO_2 uptakes where the percentage improvement of the annular pipe of aspect ratio 2.8 rises from 35.2% to 37.8%, 39.5%, and 41.8% when fins attached to the annular pipe at fin-pitching of 50 mm,

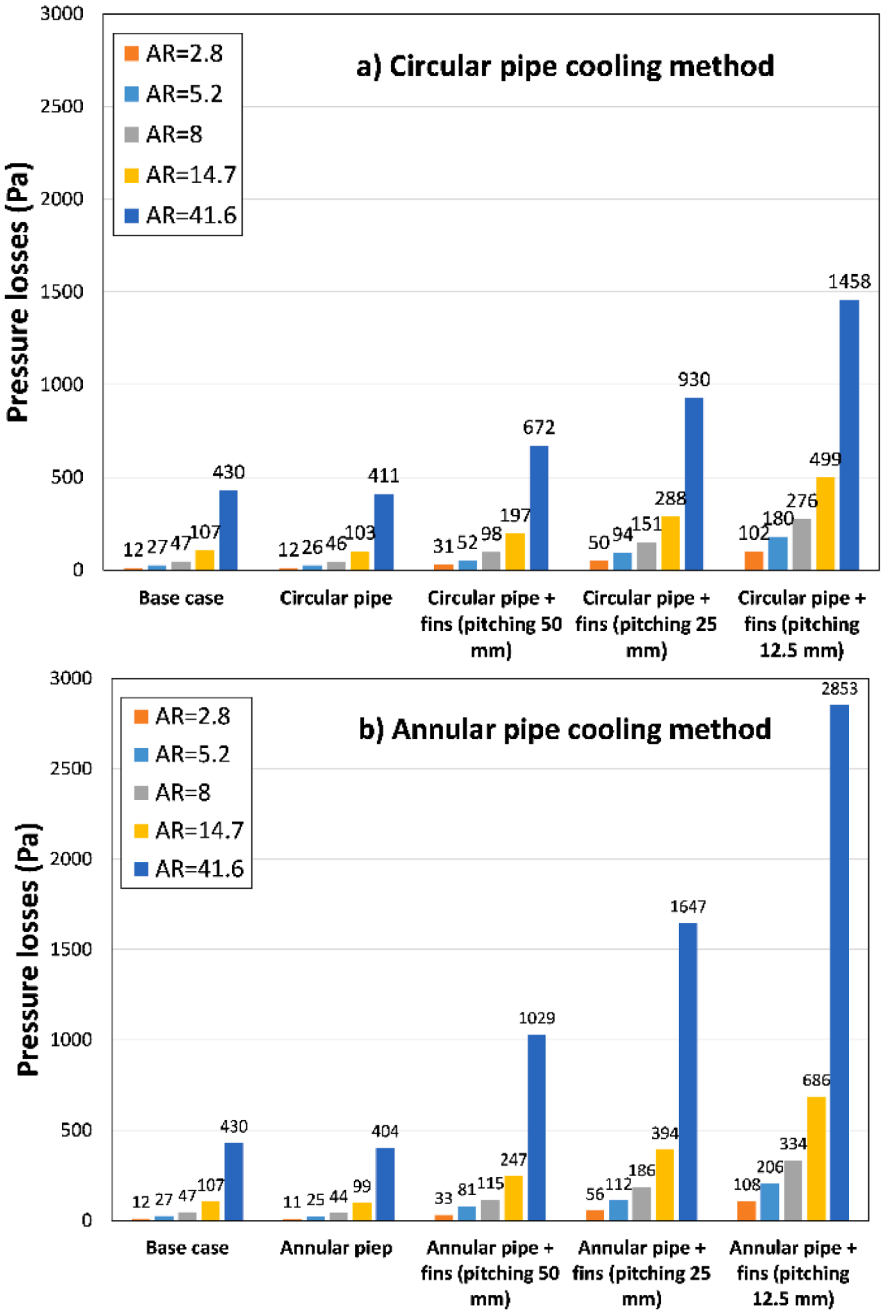


Fig. 21. Pressure losses through the adsorption bed for (a) circular pipe (b) annular pipe.

25 mm, and 12.5 mm, respectively.

Finally, it worth mentioning that the same methodology can be followed for different applications, including solar-driven adsorption chillers for air-conditioning applications and solar adsorption desalination systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. (Model validation)

The CFD model for the carbon dioxide adsorption on Mg-MOF-74 is validated in the authors' previous work [45]. Fig. S1 displays the concentration ratio curves of gas mixture (20% CO_2 and 80% N_2 vol.%) recorded at the bed outlet. The bed has a length of 7 cm and a diameter of 4 mm. The gas mixture is fed to the bed at a flow rate of 10 sccm and a temperature of 297 K. It is clear from the figure that there is an excellent agreement between the experimental data and the CFD simulation results.

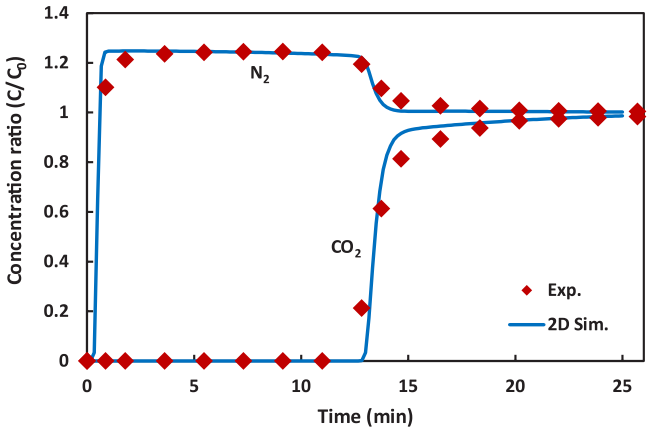


Fig. S1. Comparison of the breakthrough curves between the experimental and CFD simulation [45] (inlet conditions: P = 101.325 kPa and T = 297 K).

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